

Sebastian Caniglia

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EDUCATION

Georgia Institute of Technology

Bachelor of Science in Mathematics

Atlanta, GA

Aug. 2020 – May 2024

EXPERIENCE

Equity Trader

Trillium Trading

Sep. 2024 – Apr. 2025

Chicago, IL

- Developed and executed primarily event-driven discretionary trading strategies in U.S.-listed equities

Undergraduate Teaching Assistant

Georgia Institute of Technology

Jan. 2023 – May 2024

Atlanta, GA

- Regression and Forecasting: offered assistance to engineering students with their statistics projects in Python, R
- Linear Algebra, Multivariable Calculus: taught weekly studio sessions in foundational mathematics courses

Software Engineering Intern

Bank of America

June – Aug. 2022

Dallas, TX

- Immersed myself in the Climate and Environment Risk Data Platform, a data warehouse that collects bank asset and industry data about financed emissions to inform analytics, policy making, and investment strategy
- Developed Python scripts that perform data validation checks as part of an integrated ETL pipeline
- Incorporated the scripts into a KornShell orchestration framework

PROJECTS

Deep Music Assistant

April 2025

- Designed and developed an intelligent system that creates personalized playlist recommendations
- Interpolates structured output from Meta-Llama 3.1 70B model through AwanLLM API, persistent user storage via SQLite, and functionality with Spotify's Web API
- Implements secure OAuth flow with HTTPS server for Spotify authentication
- Utilized comprehensive error handling and modular architecture, demonstrating proficiency in API integration, database design, and secure app development
- **Key technologies:** Python, SQLite, REST APIs, OAuth, LLM Integration, Multi-threading, SSL/TLS

WHOOOP+ Interactive Dashboard

April 2025

- Built a full-stack web application that visualizes and analyzes health metrics from WHOOP wearable devices
- Leveraged Django and modern data visualization libraries (HoloViews, Panel), to feature interactive time-series analysis of heart rate variability (HRV) and resting heart rate (RHR) data with configurable moving averages
- Implemented responsive, dynamic processing and visualization updates using Pandas, NumPy and Bokeh backend for real-time user interaction
- **Key technologies:** Python, Django, Data Visualization (HoloViews, Panel, Bokeh), Data Analysis (Pandas, NumPy), Web Development, Time Series Analysis

RESEARCH

Mathematics Summer REU

School of Mathematics — Georgia Tech

May – Aug. 2023

- Conducted research project with professors Alex Blumenthal (Georgia Tech) and Evan Camrud (Colorado State)
- Studied Sobolev regularization properties of Markov transition operators by way of Fourier series representations

Predictive Analytics and Intelligent Systems Lab

H. Milton Stewart School of Industrial and Systems Engineering — Georgia Tech

Jan – May 2022

- Contributed to a multi-stage data analysis algorithm by developing a Variational Autoencoder (VAE) to process sensor data from combustion engines, enabling real-time classification and diagnostics of engine health
- Achieved diagnostic accuracy of 85% and significantly improved the pipeline's speed and reliability